

SANDEEP

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EDUCATION

B.Tech in Computer Science

Final Year

PES University, EC Campus, Bengaluru

TECHNICAL SKILLS

- **Languages:** Python, C
- **Machine Learning / NLP:** Scikit-learn (Logistic Regression, Naive Bayes, Random Forest), TF-IDF, NLTK, LIME (Explainable AI), DistilBERT, QLoRA fine-tuning, RAG pipelines (FAISS, BM25, ChromaDB), LLM API integration (Gemini, Grok, OpenAI), Prompt Engineering
- **Data Analysis:** Pandas, NumPy, Matplotlib, Jupyter Notebook, Google Colab
- **Tools & Platforms:** Git & GitHub, Linux (Fedora — kernel-level debugging), Docker, Bash scripting, Streamlit
- **Test Automation:** Playwright, Appium, YOLO, Tesseract OCR
- **Spoken Languages:** English, Hindi, Kannada, Marathi

EXPERIENCE

Independent Systems Debugging — Linux Kernel Bug Report

- Diagnosed a Bluetooth audio failure on Fedora Linux, tracing it to a PipeWire/WirePlumber/BlueZ misconfiguration on a MediaTek MT7921 chipset.
- Filed a Linux kernel bug report (kernel.org Bugzilla #221706) for a related btusb/btmtk MT7921 issue — confirmed accurate by the upstream kernel maintainer, with a fix pending in an upcoming release.

PROJECTS

AI-Assisted Test Case Generation for Web & Mobile UI

Aug 2025 – Present

Team Capstone Project (4-person team, guided by Dr. Farida Begam) — [GitHub Repo](#)

- Co-developing a dual-architecture framework that auto-generates functional test cases directly from live UI state: Architecture A (DOM / View Hierarchy — structure-based) and Architecture B (YOLO + OCR — vision-based).
- Pipeline extracts UI context, sends it to an LLM (tested with Gemini, Grok, and OpenAI APIs) to generate structured JSON test cases, then converts and executes them via Playwright (web) and Appium (mobile).
- Built and evaluated the AI model component (Feb–Mar 2026); currently reworking it to raise accuracy, and distilling a smaller open-source model to replace paid API calls for low-cost deployment.

Multi-Aspect Sentiment Analysis with Explainability

Feb – Mar 2026

Personal Project — [Live Demo](#)

- Built an end-to-end NLP pipeline classifying review sentiment with aspect-level breakdown (quality, price, size, shipping) using TF-IDF features with Logistic Regression, Naive Bayes, and Random Forest.
- Added LIME-based explainability for prediction transparency; deployed as an interactive Streamlit dashboard supporting single-review and batch CSV analysis.
- Evaluated on 100 balanced test samples (63% accuracy, 0.595 macro F1); documented class-level findings including positive-prediction bias and weak neutral-class detection.

Other ML Explorations

In Progress

- Character-Driven LLM with Long-Term Memory — fine-tuning Llama 3.2 3B with QLoRA and building a RAG-based long-term memory layer with ChromaDB, so a conversational agent can retain context across sessions.
- RAG-based Answer Engine — designing a hybrid BM25 + FAISS retrieval pipeline with DistilBERT-based intent classification and cross-encoder reranking.

CERTIFICATIONS

- [IBM SkillsBuild — Data Fundamentals](#) (March 2025)
- [IBM Skills Network — Data Science](#) (Jul 2025 – Jan 2026)
- [IBM Skills Network — Data Analysis with Python](#) (Jul 2025 – Jan 2026)
- [IBM Skills Network — Python 101 for Data Science](#) (Jul 2025 – Jan 2026)